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Generic Abstract and Generic Interface

What to Submit:

Screen shots of your output, - Done

written code, -Done

lesson learned, any known errors or bugs in your code in a Word document. -Done

In this lesson I learned about Generic Abstract and Generic Interface At first, it was very difficult for me to understand it and my Eclipse kept crashing and deleting everything. So I read a book about Java from the beginning to go through all the material and then everything connected to me I managed to understand what was happening in the code <https://tal-web.co.il/assets/docs/java.pdf> This file is in my native language

Summary of the book:

An abstract class is a parent class from which you cannot create an object directly, for example, "animal" is too general a concept. There is no such thing as just "animal". What does exist is a cat, a dog, a tiger. So the "animal" class will be abstract and the cat will inherit its properties from it. Interface is like a contract that tells the class "If you want to use me, you must implement all my functions." For example, if there is an interface of

"cat actions" that contains a function for drinking milk, every class that uses this interface must write this function. The difference between them is that an abstract class is a real father that passes attributes to a child. An interface is just a list of requirements that the class must meet.

My code:

```
package MainClass;
import java.util.ArrayList;
interface Printable{//i created an interface this forces all the classes to implement the print method
void print();//every class has to implement this method
}
//part 1 the abstract class with the T -generic
abstract class ListPrinter<T>implements Printable {
    protected ArrayList<T>list; //
    public ListPrinter(ArrayList<T> list) {
        this.list = list; // this will save it in list
    }
    public abstract void print();//this will make sure that every other class will go to print
    //here there is no implementation because each child class printers differently
}
// part 2-this is String for this "xžx—xœx$х"
class StringListPrinter extends ListPrinter<String>{
    public StringListPrinter (ArrayList<String>list) {
        super(list);//this puts it to class list
    }
    public void print() {
        for (String item : list) { // type String- loop through all items in the list and printers it
            System.out.println(item);
        }
    }
}
//part 3 - for class Integer
// same thing just with Integer
class IntegerListPrinter extends ListPrinter<Integer>{
    public IntegerListPrinter (ArrayList<Integer>list) {
        super(list);// send the list to the parent class
    }

    public void print() {
        // print all numbers in the list
        for (Integer item : list) { //type Integer
```

```

        // loop through all items in the list and printers it
        System.out.println(item);
    }
}
}

public class MainClass {
    public static void main(String[] args) {
        ArrayList<String> citiesList = new ArrayList<>();
        citiesList.add("San Francisco");
        citiesList.add("Paris");
        citiesList.add("Lima");
        citiesList.add("Saigon");
        citiesList.add("Bangkok");
        citiesList.add("Tokyo");
        citiesList.add("Cairo");
        StringListPrinter printerOne=new StringListPrinter(citiesList);
        // i create on object to print the list type String
        System.out.println("The city list:");
        printerOne.print();
        //2. Your code goes here
        // created a list of numbers
        ArrayList<Integer>numberList=new ArrayList<>(); // i create an array list
        // i create on object to print the list type Intender , i can do this because i use generics
        for (int i =1;i<=5;i++) { // for loop to make the list 1 to 5 :)
            numberList.add(i); //to add to list
        }

        IntegerListPrinter printTwo =new IntegerListPrinter(numberList);
        System.out.println("the number list:"); //print this
        printTwo.print(); // print the list of numbers
    }
}

```

Output:

```

The city list:
San Francisco
Paris
Lima
Saigon
Bangkok
Tokyo
Cairo
the number list:
1
2
3

```

4

5

